

# NMBS-C7447 at $z = 1.80$ in CPTG

Native Locking and Local Structural Invariance Compared

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## Abstract

Two Curvature Polarization Transport Gravity (CPTG) analyses have been applied independently to the same compact quiescent galaxy, NMBS-C7447 at  $z = 1.80$ . They use the same measured redshift, stellar velocity dispersion, effective radius, Sérsic profile, projected flattening, and stellar-mass normalization; they also return the same observed-epoch quantities. Here  $a_\star$  denotes the galaxy's object-dependent structural acceleration, while  $a_\pi$  denotes the native cosmological acceleration selected by redshift. Their difference appears only after the observed state has been established. The native-locking analysis holds  $a_\star/a_\pi$  constant along a native-phase comparison trajectory, which requires  $a_\star \propto E_\pi$ . The local-structural-invariance analysis instead holds the intrinsic bound-system state fixed, so  $a_\star$ ,  $R_c$ , the local field, and Structural Mode remain unchanged while  $a_\pi$  evolves. At the observed phase both analyses give  $a_\star = (1.708 \pm 0.613) \times 10^{-9} \text{ m s}^{-2}$ ,  $a_\pi = 1.64863 \times 10^{-9} \text{ m s}^{-2}$ ,  $a_\star/a_\pi = 1.036 \pm 0.372$ , and  $N = 3.06465$ . At the normalized present phase their counterfactual continuations diverge: native locking gives  $a_\star = (5.769 \pm 2.070) \times 10^{-10} \text{ m s}^{-2}$ ,  $R_c = 761.34 \pm 273.16 \text{ Gpc}$ , and an unchanged ratio  $1.036 \pm 0.372$ , whereas local structural invariance retains the observed  $a_\star$  and  $R_c$  and gives a ratio  $3.067 \pm 1.101$ . The two prescriptions are therefore distinct native-phase continuation rules anchored to the same measured state, not competing fits to the observed galaxy. NMBS-C7447 alone cannot select between them; the discriminating test requires structurally controlled systems at independent native phases, preferably with resolved multi-radius determinations of  $a_\star$ .

## 1 Purpose of the Comparison

NMBS-C7447 has been used in two separate CPTG analyses that intentionally ask different questions of the same observed galaxy. The first examines a candidate cross-scale native-locking relation between the galaxy's object-dependent structural acceleration and the native cosmological acceleration [5]. The second asks whether the bound-system structural acceleration can remain locally determined by the intrinsic galaxy state while redshift supplies only native cosmological phase [6]. Both analyses are built from the observational measurements of van de Sande et al. [1] and use the same current CPTG native branch, bound-system field equation, and Structural Mode construction [2, 3, 4].

The purpose of the present paper is not to replace either analysis or to combine their hypotheses into a third prescription. It is to place their methods side by side, identify

exactly which calculations are common, identify the first mathematical point at which they diverge, and state what the existing NMBS-C7447 observation can and cannot decide.

This comparison is useful because the two papers agree on every directly evaluated quantity at the observed epoch. Their distinct conclusions arise from different rules for changing native phase after that common state has been established. Without separating the common observed solution from the later phase continuation, the agreement at  $z = 1.80$  can be mistaken for evidence favoring one continuation over the other. The observed agreement therefore does not by itself favor either continuation.

## 2 CPTG Quantities Needed for the Comparison

Only a small part of the CPTG framework is required to follow this paper. The symbol  $a_\star$  denotes the object-dependent structural acceleration scale of the bound galaxy. It is inferred from the galaxy’s own structure and dynamics. The symbol  $a_\pi$  denotes the native cosmological acceleration associated with the redshift-selected CPTG  $\pi$ -branch phase. The corresponding quantities  $R_c$  and  $R_C$  are curvature-transport representations of those two acceleration scales; they are not the visible radius of the galaxy or a literal size of the Universe.

Two different quantities also use the letter  $N$ . The native phase coordinate is  $N_\pi = \ln a$  and tells where the observation lies on the native cosmological history. Structural Mode  $N$  is a post-solution property of the bound galaxy and describes where the radial variation of the solved CPTG field is concentrated. The logical order used throughout this paper is therefore: redshift fixes native phase and  $a_\pi$ ; galaxy observations determine  $a_\star$ ; the baryonic source is used to solve the radial field and evaluate Structural Mode; only then is a native-phase continuation rule applied. Keeping those stages separate is the central organizing principle of the comparison.

## 3 One Observed Galaxy, One Common Input Set

Both papers begin from the same source measurements:

$$\sigma_\star = 294 \pm 51 \text{ km s}^{-1}, \quad R_e = 1.64 \pm 0.15 \text{ kpc}, \quad (1)$$

with Sérsic index  $n = 5.3 \pm 0.4$ , projected axis ratio  $q_{\text{proj}} = 0.71 \pm 0.01$ , stellar mass of order  $1.5 \times 10^{11} M_\odot$ , and spectroscopic redshift  $z = 1.80$  [1]. The published dynamical mass is not treated as an independent validation datum in either CPTG paper because it is itself constructed from the measured dispersion, radius, and a Sérsic-dependent virial coefficient.

The two analyses also share the same division of observational roles. Redshift selects native cosmological phase. The stellar velocity dispersion and effective radius supply the declared one-radius pressure-support estimate of the object-dependent structural acceleration. The measured Sérsic profile, stellar mass, and flattening specify the baryonic source used in the radial field reconstruction. Structural Mode is evaluated only after the CPTG field has been solved.

The shared one-radius support estimator is

$$a_\sigma = \frac{\sigma_\star^2}{R_e}, \quad (2)$$

and both papers use it as the available estimate of the same CPTG object parameter  $a_\star$  for this unresolved pressure-supported reduction. It is not asserted to be the universal definition of  $a_\star$ . Both source analyses explicitly allow a future resolved Jeans or multi-radius reconstruction to replace this scalar estimate without changing the physical meaning of  $a_\star$  [5, 6].

| Input                         | Adopted value                     | Role common to both papers         |
|-------------------------------|-----------------------------------|------------------------------------|
| Spectroscopic redshift        | $z = 1.80$                        | Selects native $\pi$ -branch phase |
| Stellar dispersion            | $294 \pm 51 \text{ km s}^{-1}$    | Pressure-support observable        |
| Circularized effective radius | $1.64 \pm 0.15 \text{ kpc}$       | One-radius support scale           |
| Sérsic index                  | $5.3 \pm 0.4$                     | Radial source shape                |
| Projected axis ratio          | $0.71 \pm 0.01$                   | Oblate source-geometry constraint  |
| Stellar mass                  | $\sim 1.5 \times 10^{11} M_\odot$ | Radial baryonic normalization      |

Table 1: The two CPTG papers begin from the same NMBS-C7447 observations. Their methodological difference is not an input-data difference.

## 4 The Common Native-Phase Placement

This step uses the measured redshift alone to locate NMBS-C7447 on the native CPTG cosmological geometry; the galaxy’s mass, size, dispersion, and solved structural quantities do not enter the placement. Both papers use the same native CPTG  $\pi$  branch. With

$$\mathcal{B}_\pi = 1 - \frac{1}{\pi}, \quad \mathcal{T}_\pi = \frac{1}{\pi}, \quad (3)$$

and

$$E_\pi(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}, \quad (4)$$

the measured redshift selects

$$a = \frac{1}{1+z} = 0.3571428571, \quad N_\pi = \ln a = -1.029619417, \quad (5)$$

and

$$E_\pi(z = 1.80) = 2.960729196. \quad (6)$$

The common native present acceleration anchor is

$$a_{\pi,0} = \pi^{3/2} \times 10^{-10} = 5.568327997 \times 10^{-10} \text{ m s}^{-2}, \quad (7)$$

so the native acceleration at the observed phase is

$$a_\pi(z = 1.80) = a_{\pi,0} E_\pi(z) = 1.648631127 \times 10^{-9} \text{ m s}^{-2}. \quad (8)$$

The corresponding native curvature-transport radius is

$$R_C(z) = \frac{48\pi c^2}{a_\pi(z)} = 266.4149 \text{ Gpc.} \quad (9)$$

No galaxy structural quantity is used to obtain these native-phase values in either paper. Conversely, neither paper uses  $a_\pi$  to determine the observed one-radius  $a_\star$ .

## 5 The Common Observed Bound-System Solution

Using the measured dispersion and radius, both analyses obtain

$$a_\star \simeq a_\sigma = (1.708048772 \pm 0.612835118) \times 10^{-9} \text{ m s}^{-2}, \quad (10)$$

with corresponding bound-system structural curvature scale

$$R_c = \frac{48\pi c^2}{a_\star} = 257.1471 \pm 92.2625 \text{ Gpc.} \quad (11)$$

This large value is a CPTG structural curvature-transport scale associated with  $a_\star$ ; it is not the physical radius of NMBS-C7447, whose stellar body is measured in kiloparsecs.

Because the native and local curvature radii have the same numerator, both papers obtain the same dimensionless observed-state ratio,

$$\frac{R_C}{R_c} = \frac{a_\star}{a_\pi} = 1.036040594 \pm 0.371723612. \quad (12)$$

The native-locking paper denotes this ratio by  $\eta_\star$ . The local-invariance paper denotes the same algebraic ratio by  $\Xi_{\star\pi}$  because it treats the quantity as a descriptive phase coordinate rather than as a quantity constrained to remain fixed. At the observed phase there is no numerical difference between the two notations.

The central accelerations differ by only 3.60%, or approximately  $0.10\sigma$  relative to the propagated observational uncertainty. Both papers also obtain essentially the same distance-coordinate sensitivity. The acoustic and luminosity quantities used for this check are alternative distance-reporting coordinates evaluated at the same native phase; they are not additional native cosmologies. Re-expressing the source-dependent physical radius across that comparison span moves the ratio only from approximately

$$1.0073 \pm 0.3614 \quad (13)$$

to

$$1.0894 \pm 0.3909. \quad (14)$$

Thus the near-unit observed-state ratio is common to both analyses and is not generated by choosing one source-distance convention.

## 6 The Common Radial Field and Structural Mode

The one-radius estimate supplies an overall structural scale, but it does not describe how the solved field is organized across the galaxy. This section asks that second question. The radial reconstruction is not used to refit the observed one-radius  $a_*$ ; it uses the common observed source and structural scale to determine where the returned CPTG field changes most strongly. Structural Mode  $N$  is the downstream readout of that radial organization and is distinct from the native phase coordinate  $N_\pi$ .

Both papers extend the one-radius support calculation by reconstructing the stellar source from the measured Sérsic profile, assuming a spatially constant stellar mass-to-light ratio and normalizing the profile to the reported stellar mass. Both then solve the same registered CPTG radial response,

$$g = g_{\text{bar}} + \frac{\sqrt{a_* g_{\text{bar}}} + g_{\text{geom}}}{[1 + (g/a_*)^{123/50}]^{11/50}}, \quad (15)$$

with

$$g_{\text{geom}}(r) = \frac{1}{2} a_* \frac{(r/R_c)^{7/5}}{1 + (r/R_c)^{12/5}}. \quad (16)$$

Structural Mode is then evaluated from the solved field through

$$C_g(r) = g(r)^2 + \kappa_{\text{struct}}^2 \left( \frac{dg}{dr} \right)^2, \quad \kappa_{\text{struct}} = 1 \text{ kpc}, \quad (17)$$

$$w(r) = \left| \frac{dC_g}{dr} \right|, \quad (18)$$

and

$$\lambda = \frac{\int_{R/5}^R r w(r) dr}{\int_{R/5}^R w(r) dr}, \quad N = \frac{R}{\lambda}. \quad (19)$$

For the measured-flattening edge-on source, both papers return

$$N_{\text{obl}} = 3.06465. \quad (20)$$

They also report the same source and geometry sensitivities: one-at-a-time source perturbations span approximately 3.05339–3.08142, the simultaneous measured-flattening source corner spans 3.03026–3.13356, and the central near-thin geometry sequence approaches  $N \simeq 3.1159$  [5, 6].

The agreement is expected. Up to this point the two papers have used the same observed source, the same observed one-radius  $a_*$ , and the same radial field equation. Structural Mode therefore does not discriminate between the two phase prescriptions at the actual observed epoch.

## 7 Where the Two Methods First Diverge

The methods separate only when the observed state is represented at a different native phase. The derivatives below describe counterfactual comparison trajectories with respect to the native phase coordinate  $N_\pi$ ; they are not equations for the literal time evolution of NMBS-C7447. The distinction can be written directly from quantities already used in both papers. Since

$$\frac{a_\star}{a_\pi} = \frac{a_\star}{a_{\pi,0} E_\pi(a)}, \quad (21)$$

differentiation with respect to  $N_\pi = \ln a$  gives

$$\frac{d \ln(a_\star/a_\pi)}{dN_\pi} = \frac{d \ln a_\star}{dN_\pi} + \frac{\pi}{2} \frac{\mathcal{T}_\pi a^{-\pi}}{\mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}}. \quad (22)$$

At  $z = 1.80$ , the native-branch term in Eq. (22) is

$$1.448641818. \quad (23)$$

Equation (22) exposes the two papers as distinct choices for which side of the comparison is held invariant.

### 7.1 Native-locking continuation

The native-locking paper imposes

$$\frac{d \ln(a_\star/a_\pi)}{dN_\pi} = 0. \quad (24)$$

Therefore

$$\frac{d \ln a_\star}{dN_\pi} = -\frac{\pi}{2} \frac{\mathcal{T}_\pi a^{-\pi}}{\mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}} = \frac{d \ln E_\pi}{dN_\pi}, \quad (25)$$

which integrates to

$$a_\star(a) = \eta_\star a_{\pi,0} E_\pi(a). \quad (26)$$

The invariant quantity is the cross-scale ratio. Consequently  $a_\star$  and  $R_c$  change along the comparison trajectory as native phase changes.

### 7.2 Local-structural-invariance continuation

The local-invariance paper instead fixes the intrinsic bound-system state  $\mathcal{S}$  and imposes

$$\left. \frac{\partial a_\star}{\partial N_\pi} \right|_{\mathcal{S}} = 0. \quad (27)$$

Therefore

$$\left. \frac{d \ln(a_*/a_\pi)}{dN_\pi} \right|_{\mathcal{S}} = \frac{\pi}{2} \frac{\mathcal{T}_\pi a^{-\pi}}{\mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}}, \quad (28)$$

and at the observed phase

$$\left. \frac{d \ln(a_*/a_\pi)}{dN_\pi} \right|_{z=1.80, \mathcal{S}} = 1.448641818. \quad (29)$$

The invariant quantities are the local structural scale and the full local field constructed from that fixed state. The cross-scale ratio changes because  $a_\pi$  changes.

### 7.3 Why the two continuations are mathematically distinct

The distinction is not semantic. If  $a_*/a_\pi$  is constant while  $a_\pi$  changes, then  $a_*$  must change. If the intrinsic state is fixed in the local-invariance sense, then  $a_*$  does not change. Both conditions could hold over a finite phase interval only if  $a_\pi$  itself were phase independent over that interval, which the native  $\pi$  branch is not.

Accordingly, the native-locking continuation is best read as a cross-scale comparison trajectory anchored to the observed galaxy, whereas the local-invariance continuation is a strict fixed-intrinsic-state phase relocation. They begin from the same measured state but do not represent the same counterfactual operation after leaving the observed phase.

## 8 Numerical Divergence Away from the Observed Phase

In plain terms, native locking changes the galaxy-side structural scale so that the cross-scale ratio is preserved, whereas local structural invariance preserves the galaxy-side structural scale and allows the ratio to change. The difference becomes especially transparent at the normalized native present  $a = 1$ . The native-locking paper preserves the observed ratio and therefore carries the structural acceleration to

$$a_{*,\text{lock}}(1) = \frac{a_{*,\text{obs}}}{E_\pi(a_{\text{obs}})} = (5.769 \pm 2.070) \times 10^{-10} \text{ m s}^{-2}, \quad (30)$$

with

$$R_{c,\text{lock}}(1) = 761.34 \pm 273.16 \text{ Gpc}, \quad (31)$$

and

$$\left( \frac{a_*}{a_\pi} \right)_{\text{lock},1} = 1.036 \pm 0.372. \quad (32)$$

The local-invariance paper instead retains

$$a_{*,\text{local}}(1) = (1.708 \pm 0.613) \times 10^{-9} \text{ m s}^{-2}, \quad (33)$$

$$R_{c,\text{local}}(1) = 257.15 \pm 92.26 \text{ Gpc}, \quad (34)$$

while the ratio evolves to

$$\left(\frac{a_\star}{a_\pi}\right)_{\text{local},1} = 3.067 \pm 1.101. \quad (35)$$

The factor separating the two present-coordinate structural accelerations is exactly the native phase factor between the observed epoch and  $a = 1$ :

$$\frac{a_{\star,\text{local}}(1)}{a_{\star,\text{lock}}(1)} = E_\pi(z = 1.80) = 2.960729196. \quad (36)$$

The same factor appears inversely in the corresponding structural curvature radii.

| Quantity                                      | Native locking                                      | Local structural invariance         |
|-----------------------------------------------|-----------------------------------------------------|-------------------------------------|
| Observed $a_\star$                            | $(1.708 \pm 0.613) \times 10^{-9} \text{ m s}^{-2}$ |                                     |
| Observed $R_c$                                | $257.15 \pm 92.26 \text{ Gpc}$                      |                                     |
| Observed $a_\star/a_\pi$                      | $1.036 \pm 0.372$                                   |                                     |
| Observed $N_{\text{obl}}$                     | $3.06465$                                           |                                     |
| Phase invariant quantity                      | $a_\star/a_\pi$                                     | intrinsic local state and $a_\star$ |
| $d \ln a_\star / dN_\pi$ at $z = 1.80$        | $-1.44864$                                          | $0$                                 |
| $d \ln(a_\star/a_\pi) / dN_\pi$ at $z = 1.80$ | $0$                                                 | $+1.44864$                          |
| $a_\star$ at $a = 1$                          | $(5.769 \pm 2.070) \times 10^{-10}$                 | $(1.708 \pm 0.613) \times 10^{-9}$  |
| $R_c$ at $a = 1$                              | $761.34 \pm 273.16 \text{ Gpc}$                     | $257.15 \pm 92.26 \text{ Gpc}$      |
| $a_\star/a_\pi$ at $a = 1$                    | $1.036 \pm 0.372$                                   | $3.067 \pm 1.101$                   |

Table 2: Observed-state agreement and native-phase divergence. Values at  $a = 1$  are comparison coordinates anchored to the measured  $z = 1.80$  state, not forecasts of the galaxy’s literal future evolution.

## 9 What Happens to Structural Mode

At the observed phase the Structural Mode calculation is identical in the two papers because the source and  $a_\star$  supplied to the radial field solve are identical. The result is therefore the same:

$$N_{\text{obl}} = 3.06465. \quad (37)$$

The local-invariance paper goes further and performs an explicit fixed-state phase sweep. Because  $a_\star$ ,  $R_c$ , the baryonic source, and the solved field remain fixed, it obtains

$$\left.\frac{\partial g(r)}{\partial N_\pi}\right|_{\mathcal{S}} = 0, \quad \left.\frac{\partial N}{\partial N_\pi}\right|_{\mathcal{S}} = 0, \quad (38)$$

and the numerical phase-sweep spread in  $N$  is zero to stored precision [6].

The native-locking paper does not make that fixed-field statement. Its phase continuation requires  $a_\star$  and  $R_c$  to change in order to preserve  $a_\star/a_\pi$ . Since those quantities enter



the radial CPTG response equation, a native-locking continuation is not simultaneously the fixed local-field continuation used in the local-invariance paper. The observed value  $N = 3.06465$  therefore remains a common measured-state result, but the two papers do not imply the same off-epoch behavior for the solved local field.

This distinction also prevents Structural Mode from serving as a retrospective tie-breaker for NMBS-C7447 itself. The two analyses have already returned the same  $N$  from the same observed field. A discriminating use of  $N$  must occur in a population in which source structure, support, and native phase vary independently enough to test the competing continuation rules.

## 10 Why the Near-Unit Observed Ratio Does Not Select a Method

The observed ratio

$$\frac{a_\star}{a_\pi} = 1.036 \pm 0.372 \quad (39)$$

is one of the most visually striking numbers in both papers. In the native-locking analysis it is naturally compared with the strongest unit-lock benchmark. In the local-invariance analysis it is simply the observed cross-scale location of the galaxy, with no preferred value imposed.

The comparison here shows why that difference of interpretation matters. Both papers obtain the ratio only after using the same redshift and the same one-radius pressure-support estimate. Therefore the numerical proximity to unity is not an independent result available to one paper but absent from the other. It is a shared observed-state datum.

Native locking gains empirical content only when the ratio is tested for persistence across independent phases or systems after structural differences are controlled. Local structural invariance gains empirical content only when  $a_\star$  is tested for the absence of a residual phase dependence after the bound-system state is controlled. The single NMBS-C7447 observation supplies the anchor for both tests but cannot measure either derivative directly.

The source uncertainty reinforces this conclusion. The approximately 36% uncertainty of the one-radius  $a_\star$  estimate is more than ten times the central 3.60% offset from unity. The existing object is therefore consistent with unit locking, but it is also fully compatible with the local-invariance interpretation in which the same near-unit value has no privileged status.

## 11 Shared Methodological Limitation: One-Radius Inference of $a_\star$

The principal observational limitation common to both analyses is the information content of the available galaxy kinematics. Both papers estimate  $a_\star$  from the integrated pressure-support coordinate  $\sigma_\star^2/R_e$ . That reduction is appropriate as a one-radius summary when the available observation provides an integrated dispersion and effective radius, but it cannot contain the same information as a resolved multi-radius dynamical reconstruction.

This matters for the comparison because the competing phase prescriptions act on  $a_*$ . A future test should therefore avoid making the quality of the phase test depend unnecessarily on a scalar approximation when resolved kinematics are available. The preferred hierarchy is observational rather than definitional: a resolved multi-radius solution should supersede a one-radius estimator as the measurement of the same underlying  $a_*$  whenever the data support it.

The two NMBS papers already preserve that distinction. Neither identifies  $a_*$  universally with  $\sigma^2/R$ . The comparative conclusion is therefore conservative: the present disagreement concerns the phase continuation of the same object-dependent structural quantity, not competing definitions of that quantity.

## 12 From the Trajectory Difference to a Population Test

The clean mathematical distinction is first stated for one common structural anchor represented at two native phases,  $a_1$  and  $a_2$ . This is the counterfactual operation compared by the two papers; it is not yet a claim about an arbitrary pair of different galaxies.

Under strict local structural invariance of that same intrinsic state,

$$\frac{a_*(a_2)}{a_*(a_1)} = 1. \quad (40)$$

Under native locking,

$$\frac{a_*(a_2)}{a_*(a_1)} = \frac{E_\pi(a_2)}{E_\pi(a_1)}. \quad (41)$$

Equations (40) and (41) are trajectory statements for that common anchor. They provide the cleanest mathematical comparison because they use quantities already present in the two papers and make no appeal to the accidental closeness of one observed ratio to unity.

Real galaxies do not provide repeated observations of a literally frozen intrinsic state across cosmic time. A population test can only approximate the comparison after controlling the structural information that both papers identify as relevant: stellar mass profile, size, geometry, pressure or mixed support, and post-solution radial organization. Structural Mode can contribute to that control but should not substitute for the full source and support state. Wherever possible, the structural acceleration should be inferred from resolved multi-radius dynamics rather than from a one-radius proxy.

The decisive question is then residual. After the observable bound-system differences have been accounted for, does the inferred  $a_*$  require the native phase factor  $E_\pi$ , or does it remain locally determined without that additional phase term? That is the empirical distinction between the two methods.

### 13 Interpretive Boundaries

Several limits are common to both source papers and remain in force here.

First, neither phase continuation is a literal evolutionary history of NMBS-C7447. A real galaxy can change its mass profile, size, orbital support, geometry, and environment. Both source papers explicitly distinguish their comparison trajectories from forecasts of the galaxy’s physical future.

Second, the source study’s standard cosmological distance convention is retained only where needed to respect how the published physical radius and stellar mass were reported. The governing phase in both CPTG papers is the native  $\pi$  branch; the acoustic and luminosity quantities discussed above remain distance-coordinate sensitivity tests at that same phase.

Third, the two papers do not constitute statistically independent observational tests of the observed values  $a_\star$ ,  $a_\pi$ ,  $a_\star/a_\pi$ , and  $N$ . They deliberately reuse the same galaxy and much of the same numerical machinery to isolate one conceptual difference. The scientific value of the comparison is therefore mathematical and methodological: it reveals what each continuation assumes and identifies the future observations capable of distinguishing them.

Fourth, this comparison does not introduce a third interpolation between native locking and local invariance. The existing data do not justify such an interpolation. The appropriate next step is empirical discrimination, not an additional adjustable phase law.

### 14 Conclusion

The two NMBS-C7447 analyses share the same observational reduction and the same solved state at  $z = 1.80$ . Both give

$$\begin{aligned} a_\star &= (1.708 \pm 0.613) \times 10^{-9} \text{ m s}^{-2}, \\ a_\pi &= 1.64863 \times 10^{-9} \text{ m s}^{-2}, \\ \frac{a_\star}{a_\pi} &= 1.036 \pm 0.372, \quad N_{\text{obl}} = 3.06465. \end{aligned} \tag{42}$$

Their disagreement therefore does not concern the measured galaxy. It begins only when that common anchor is represented away from the observed native phase.

Native locking preserves  $a_\star/a_\pi$  and therefore requires  $a_\star \propto E_\pi$ . Local structural invariance preserves the intrinsic bound-system state and  $a_\star$ , so the ratio instead evolves as  $E_\pi^{-1}$ . At  $z = 1.80$  these choices correspond to

$$\begin{aligned} \text{native locking: } \frac{d \ln a_\star}{d N_\pi} &= -1.44864, & \frac{d \ln(a_\star/a_\pi)}{d N_\pi} &= 0, \\ \text{local invariance: } \frac{d \ln a_\star}{d N_\pi} &= 0, & \frac{d \ln(a_\star/a_\pi)}{d N_\pi} &= +1.44864. \end{aligned} \tag{43}$$

The normalized-present comparison expresses the same difference numerically: native locking carries  $a_\star$  to  $(5.769 \pm 2.070) \times 10^{-10} \text{ m s}^{-2}$  while preserving the observed ratio, whereas local invariance retains the measured  $a_\star$  and carries the ratio to  $3.067 \pm 1.101$ .

NMBS-C7447 alone cannot select between the two phase prescriptions because it supplies one observed structural state at one native phase. The decisive test is a structurally controlled population spanning independent native phases, with resolved multi-radius determinations of the same object-dependent  $a_\star$  wherever possible. Until such a test is performed, native locking and local structural invariance should remain separate, explicit CPTG hypotheses.

## Data, Code, and Evidence Availability

The observational authority common to both analyses is van de Sande et al. [1]. The native branch, structural-acceleration hierarchy, and local/native curvature-radius relations follow the current CPTG unified framework and  $\pi$ -branch comparison-coordinate paper [2, 3]. Structural Mode follows the current CPTG Structural Mode validation [4].

This comparative manuscript introduces no new observational dataset, no new galaxy fit, and no new radial field solve. Its numerical statements are algebraic comparisons of results already archived by the two source analyses [5, 6]. Their exact numerical evidence authorities are:

CPTG\_NMBS-C7447\_NativeLock\_Structural\_Evidence\_20260904\_r04\_FINAL\_EVIDENCE.zip

SHA-256: fc3076d3ee5a83fd17f1a1a75fe73844bb36974440a0a204d7009c446247420a

CPTG\_NMBS-C7447\_LocalStructuralInvariance\_Evidence\_20260904\_r01\_FINAL\_EVIDENCE.zip

SHA-256: 875c079e44e6ce4e5963987591bd7dfc8e4b7d36ecee046ab45b743702b44a2

The public CPTG source repository is available through the supporting resource [7].

## References

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